

Commonwealth Games Medalists

Power BI Dashboard

Data Analysis and Visualization Project

Project Overview

This project presents an interactive Microsoft Power BI dashboard for analyzing Commonwealth Games medalists and medal performance across countries, athletes, events, and historical years. The dashboard transforms medal data into interactive visualizations that allow users to explore country performance, athlete achievements, event-level statistics, and historical medal trends.

The analysis covers Commonwealth Games medal data from 1930 onwards, depending on the availability of records in the dataset.

Project Objectives

- Analyze medal performance across Commonwealth Games.
- Compare medal performance between countries.
- Identify top-performing athletes.
- Analyze medals across different events.
- Examine historical medal trends.
- Provide interactive filtering and exploration.
- Present the results through an intuitive Power BI dashboard.

Tools and Technologies

Tool / Technology	Purpose
Microsoft Power BI	Dashboard development and data visualization
Power Query	Data cleaning and transformation
DAX	Measures and analytical calculations
GitHub	Project documentation and version control

Dashboard Pages

1. Overview

The Overview page provides a high-level summary of Commonwealth Games medal performance.

- KPI cards
- Medal distribution analysis
- Country comparisons
- Interactive slicers
- Summary charts

- Overall medal statistics

2. Country Performance

The Country Performance page focuses on comparing medal performance between countries.

- Country-level filtering
- Gold, silver, and bronze medal analysis
- Medal distribution
- Historical country performance
- Geographic visualization
- Country comparison tables

3. Athletes and Events

The Athletes and Events page provides detailed analysis of athletes and sporting events.

- Athlete filtering
- Event analysis
- Medal counts by athlete
- Top-performing athletes
- Event-level medal analysis
- Detailed data tables

4. Historical Trends

The Historical Trends page examines changes in Commonwealth Games medal performance over time.

- Year-based filtering
- Historical medal trends
- Medal comparisons across years
- Trend charts
- Historical summary tables

Key Dashboard Features

Interactive Filters

The dashboard provides interactive slicers for:

- Country
- Year
- Athlete
- Event
- Medal type

KPI Cards

KPI cards provide a quick overview of important numerical indicators and allow users to understand overall medal statistics at a glance.

Visualizations

- Bar charts
- Column charts
- Line charts
- Area charts
- Pie and donut charts
- Tables
- Maps
- KPI cards

Cross-Filtering

The dashboard supports interactive cross-filtering. Selecting values in one visual dynamically updates related visualizations, allowing users to explore relationships within the dataset.

Data Preparation

The dataset was prepared before visualization using Power Query and Power BI. The preparation process included:

- Importing the dataset.
- Checking and correcting data types.
- Cleaning inconsistent values.
- Handling missing or unnecessary values.
- Transforming columns where required.
- Preparing categorical and numerical fields.
- Creating relationships and analytical fields.
- Developing DAX measures.

DAX and Calculations

DAX measures were used to support the dashboard analysis, including calculations related to total medals, gold medals, silver medals, bronze medals, country performance, athlete performance, and historical medal trends.

Questions Explored

- Which countries have won the most medals?
- Which countries have performed consistently over time?
- Which athletes have achieved the highest medal counts?
- How are gold, silver, and bronze medals distributed?
- Which events contribute most to a country's medal performance?
- How has medal performance changed throughout Commonwealth Games history?
- Which countries dominate particular periods or events?

Repository Structure

```
Commonwealth-Games-Medalists-PowerBI/  
|  
|-- PBI.pbix  
|-- README.md  
|  
`-- images/  
    |-- overview.png  
    |-- country-performance.png  
    |-- athletes-events.png  
    `-- historical-trends.png
```

How to Use

Step 1 — Clone the Repository

```
git clone https://github.com/your-username/  
Commonwealth-Games-Medalists-PowerBI.git
```

Step 2 — Open the Power BI File

Open **PBI.pbix** using Microsoft Power BI Desktop.

Step 3 — Explore the Dashboard

Use the available slicers, charts, tables, maps, and KPI cards to interactively explore the Commonwealth Games medal data.

Skills Demonstrated

- Data cleaning
- Data transformation
- Data analysis
- DAX
- Data visualization
- Dashboard design
- Interactive reporting
- Data storytelling
- Power BI

Future Improvements

- Adding advanced DAX measures.
- Adding additional performance indicators.
- Implementing drill-through pages.
- Adding more detailed athlete-level analysis.
- Incorporating statistical or predictive analysis.
- Improving mobile dashboard layouts.
- Adding automated data refresh functionality.
- Expanding the dataset with additional Commonwealth Games data.

Author

Gayana

Undergraduate Student
Faculty of Science
University of Peradeniya, Sri Lanka

License

This project is developed for educational and portfolio purposes. Please refer to the original dataset source for information regarding data ownership, licensing, and permitted usage.

Conclusion

The Commonwealth Games Medalists Power BI Dashboard provides an interactive platform for exploring medal performance across countries, athletes, events, and historical years. The project demonstrates how data preparation, DAX calculations, and interactive visualization can be combined in Power BI to transform raw data into meaningful analytical insights.

Built with Microsoft Power BI

Data Analytics and Visualization Project